

Institute of Actuaries of India

Subject SP4 – Pensions & Other Benefits

March 2022 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) Design features:**

- Immediate eligibility - Allow participants to enroll immediately or soon after employment
- Automatic enrollment - Auto-enroll employees at an initial percentage designed to achieve the plan's savings and income replacement goals.
- Additional matching contributions - Provide an additional matching contribution in addition to the mandatory employer contribution
- Re-enrollment - Expand the benefits of automatic features by implementing re-enrollment strategies on a periodic basis
- Discourage early withdrawals and loans from retirement savings - Set limits on outstanding loan times, waiting periods
- Encourage terminated members to leave balances in the plan or to roll them over to an alternative retirement plan.
- Provide limited menu of investment options - Enough options to cover basic asset classes or a default option
- Generous employer contributions - Helps accumulate higher retirement assets
- Provide a guaranteed minimum rate of return on assets

[½ mark for each point, Max 6]

ii) The company could institute a DC plan where benefits and/or vesting could be tied to company performance. [½]

As an example, the annual allocation to the DC plan could be a percentage of pay based on sales or revenue growth relative to target. In addition, vesting of the DC contributions could be variable, based on revenue growth relative to target. [1]

Both of these options align rewards from the DC plan to company performance. [½]

Alternatively, the Company could institute a holdback plan where benefits are subject to verification of business results. [1]

A holdback provision could be implemented such that the contributions to the plan would not vest for a period of few years so that revenue can be verified. This rewards long-term sustained performance rather than short term results that could be manipulated. [1]

[Max 4]**iii) Considerations for excluding variable pay:**

- Excluding variable pay could significantly reduce the cost of the plan [½]
- Excluding variable pay could significantly reduce cost volatility [½]
- Excluding variable pay would make providing retirement estimates/administering the plan easier [½]
- Money could be spent on other programs that have a better internal ROI for the company [½]

Considerations for including variable pay:

- Including variable pay would result in more adequate retirement benefits for their employees and better tie retirement income to gross working income [½]
- Variable pay may be included in order to be competitive in the Company's industry [½]

- Including variable pay would increase the retirement benefit and may act as a retention tool for executives [½]
- If variable pay is related to performance, including variable pay aligns with the company philosophy of tying rewards to performance [½]

[Max 4]

- iv)** Invest the assets, leave the principal intact [1]
- Spend investment earnings [½]
 - Spend interest and dividends [½]
 - Realized capital gains are typically reinvested, but available to be spent [½]
 - Not guaranteed but designed to make money last for life [½]

Systematic withdrawals [1]

- Invest the assets and withdraw the principal and investment earnings with a formal method [½]

Annuity purchase

- Transfer savings to insurance company to guarantee a lifetime retirement income [½]
- Survivor benefits and guarantee period can be added [½]
- Variable annuity [½]
- Return of capital guarantee on death [½]

[Max 6]

- v)** Investment earnings
- Never run out of money
 - Entire capital available as inheritance
 - Full access to savings
 - Full control over investments
 - Can purchase annuity later

Systematic withdrawals

- Can use withdrawal strategy to ensure income available for all years
- Full access to savings
- Withdrawal and Investment control
- High returns lead to higher income
- Balance to heirs
- Can purchase annuity later

Annuity purchase

- Lifetime income guaranteed
- Can add guarantee or survivor benefit
- Post-retirement increases if indexing purchased
- Not required to invest funds
- Protection against investment volatility
- Protection against fraud
- Possible Government protection against insurer bankruptcy

- Higher income than systematic
- For Variable annuity, Benefit can increase over time

[½ mark for each point, max 6]

[26 Marks]

Solution 2:**i) Advantages**

- It produces less administration for short stayers who never become eligible for membership. [½]
- It will save the employer money, as all members build up less service in the scheme. [½]
- It can help the employer to meet their aims, e.g. target resources to long-term staff. [½]

Disadvantages

- Those in greatest need of the death-in-service benefit, e.g. young people with family responsibilities, may not be covered. [½]
- When employees are unable to join the pension scheme, they may be persuaded to make their own individual arrangements and thus be less likely to join the company scheme when they are eligible. This may not serve their best interests. [½]
- It may be more difficult to persuade someone to join the company if they have to wait to join the pension scheme. [½]
- Where the scheme requires employees to contribute, they are less likely to join after a waiting period since they will have become used to a certain level of net income. [½]

[Max 4]**ii) If the scheme was non-contributory, then compared with a contributory scheme the cost to the sponsor would be higher [1]**

... or the benefits would be less generous, if the cost to the sponsor were unchanged. [½]

Employee contributions should not be:

- too low – otherwise deductions, with the associated costs of administration, would not be worthwhile [½]
- too high – unless the package of benefits is exceptionally generous. [½]

A key factor will be the level of member contributions required by schemes operated by competitors. [1]

In particular, comparison should be made with those:

- in the same industry [½]
- for the same type of employees [½]

The employer's intentions and aims for the scheme are extremely important. [½]

Virtually all members will opt to join a non-contributory scheme. Depends on whether the sponsor paternalistic [½]

Alternatively, the sponsor may wish to promote the concept of sharing responsibilities with the employees, i.e. self-empowerment and independence rather than reliance on the sponsor. [½]

A non-contributory scheme will be simpler to administer. [½]

In the event of a surplus arising in a defined benefit scheme, the sponsor's claim to ownership of the surplus is stronger for a non-contributory scheme. [½]

DC arrangements are more likely to be contributory, since the member's fund is demonstrably the member's own property. [½]

The decision may depend on the availability and scope of tax concessions on contributions from either the sponsor or the members. [½]

[Max 7]

- iii) To an extent the trustee has a point, i.e. if you are single at death then no spouse's pension will be paid and therefore the overall benefits are lower than for someone married at death. [½]

However, a major flaw in their argument is that marital status does not stay constant, i.e. single members now may be married at death and vice-versa. [1]

It would be very unusual for a scheme to have different contribution rates for single and married members. [½]

Also, the definition of "married" may not be clear cut. Would members be classified as "married" if they are in a long-term common-law relationship? [½]

The administration of the scheme will become more complicated which will add to the cost. [½]

It will be an unpopular move with most of the scheme members as it is probable that most of them will be married. [½]

A feature of defined benefit scheme is that there are cross-subsidies. Other cross subsidies occur, e.g. between young and old, it would be too complicated to allow for all of them. [½]

[Max 4]

iv) *Liability profile*

The average term of the liabilities will be very long when the scheme begins. The term of appropriate assets will reduce over time. [½]

The nature of the benefits is also important in assessing the suitability of the assets. [½]

Early on, the liabilities will mainly be linked to earnings. As the scheme matures, more benefits may be linked to price inflation or may have fixed increases, depending on the nature of the benefits provided. [½]

Fund size

Early on there will be limited funds available and so some investment classes may not be suitable because of their size, eg direct property investment. In the longer term, new classes may become available as the scheme grows. [½]

Level of risk / security

The views of the trustees and the sponsor should be sought so as to ascertain their attitude to risk. [½]

The strength of the sponsor covenant should also be assessed. [½]

A more aggressive investment strategy could be adopted to try to increase returns and hence reduce costs [½]

The downside is that there is a risk of a significant deficit if experience turns out to be worse than expected, resulting in increased costs and a volatile contribution rate. [½]

The scheme could minimise the probability of deficit by holding assets that closely match the liabilities.

However, the cost of doing so may be unacceptably high in terms of lower investment returns ... [½]

... furthermore, surpluses are less likely to arise and hence members are less likely to be granted benefit improvements. [½]

The consequences of poor investment performance are less significant for an immature scheme ... [½]

... it may therefore wish to follow a policy of deliberately mismatching, in order to benefit from higher expected returns from assets that do not closely match the liabilities. [½]

The scheme may consider investing a greater proportion of the scheme in equities, which historically have outperformed government bonds over the long term, but have extra risks associated with them, including default and volatility. [½]

Funding level

The more healthy the funding position, the greater the level of investment freedom, increasing the scope to mismatch. [½]

Liquidity

For an immature scheme, benefit outgo is low and therefore can be paid out of contribution income. [½]

Liquidity will therefore not be a major issue and the scheme will have less need to invest in marketable and high yielding assets. [½]

Consideration should be given to any protection benefits offered, for example a lump sum payment on death in service could lead to liquidity risk for an immature scheme with only a small fund, which may lead to insurance being used. [½]

Diversification

A diversified portfolio of assets reduces risk. The optimum position between diversification and minimising investment costs needs to be sought. [½]

Restrictions / legislation

The State may restrict the types of investments held. For example, it may restrict the level of self-investment or set levels for minimum / maximum holdings of different types of investment. [½]

Tax position

The tax implications of holding certain investments should be considered. The return net of tax is the key issue. [½]

Investment expenses

The expenses associated with investment should also be considered. The return net of expenses is the key issue. [½]

Ethical / moral issues

The trustees and/or sponsor may choose to avoid certain investments from an ethical / moral viewpoint. [½]

Competitor schemes / benchmarking

The scheme is unlikely to want to risk poor performance compared to its peers, and may want to specify that the managers should not deviate significantly from the investment strategy followed by competitors' schemes [½]

[Max 9]**[24 Marks]****Solution 3:**

- i) The actual experience of a scheme should be monitored to check whether the method and assumptions adopted for financing the benefits continue to be appropriate and, if not, what changes should be made in order to achieve the desired level of prudence. The three common reasons are:
- By analysing the actual mortality experience of the pensioners, a sponsor may understand the relative importance of the said factor that influence the actual cost of the benefits. [1]
 - This knowledge will assist in future financial planning and risk management, *eg* setting terms for options, reducing liquidity risk and making decisions about the use of insurance e.g. for buying annuity for pensioners. [1]
 - Monitoring experience will also assist the sponsor in obtaining a realistic estimate of the cost of the benefit provision and help in setting valuation assumptions. [1]

[Max 3]

ii) The steps involved in the analysis are:

- Collect all relevant data at a specific date, *eg* previous valuation date. [1]
- Split the data into homogeneous groups (as far as possible), *eg* by age, sex, member/beneficiaries, mode of retirement. [1]
- The extent to which this can be carried out will be limited by the volume of data available. [1]
- This may be done here, especially being a large scheme. [½]
- Calculate the exposed to risk, and derive rates of mortality experienced using the actual number of deaths. [1]
- However, if there is insufficient credible data then the alternative approach is to:
 - group the data by age and sex (and any other splits that does not produce statistically unreliable data) [½]
 - calculate the expected number of deaths from a standard mortality table [½]
 - compare with the actual number of deaths experienced at each age. [½]
- This may be more appropriate for smaller schemes to work out any adjustment to the standard table. [½]

[Max 5]

iii)

- The data may reflect unusual experience especially the effects of particularly medical pandemic Covid-19. [½]

- The increase in the mortality rates due to Covid-19 are actually the random fluctuations in experience for even the large schemes and hence taking that into account for setting the assumptions may not reflect the true picture. [1]
- The data will cover a historic period and will not reflect any subsequent changes in life expectancy [½]
- Particularly at older ages where there are likely to be fewer members [½]
- And at younger ages where there are likely to be fewer deaths [½]
- Particularly as only one year's worth of experience has been used [½]
- There may be calculation error or errors in the data provided by the scheme [½]
- The approach omits any deaths that are "incurred but not reported" [½]
- The approach may be unsuitable for some groups of members e.g. senior executives who might be expected to display higher life expectancies [½]
- And it would be usual to determine different tables for males and females [½]
- Current experience may be helpful to assess the base table but will not provide any information on likely improvements in the longer term [½]
- Use of a base table with no allowance for future improvements will materially understate actual life expectancies [½]
- And the value placed on pension liabilities [½]
- The results may not be relevant to different cohorts with the mix of members changing over time [½]
- It is important to use assumptions appropriate to the purpose and type of valuation [½]
- For example, a funding valuation may require a prudent mortality assumption [½]
- Consider consistency with the previous valuation and other valuations (e.g. solvency) [½]

[Max 4]

iv)

- The results should be smoothed to eliminate random fluctuations such as recent spike due to Covid-19, as stated in the question [½]
- Using a longer period than one year to increase the amount of data and reduce volatility [½]
- Consideration should be given to whether there are homogeneous groups within the population with different mortality profiles [½]
- And the study undertaken for each of these groups (assuming that there is sufficient reliable information) [½]
- Particularly males/females, white collar/blue collar and potentially by other worker types (e.g. those in hazardous occupations) [½]
- If the scheme is big enough, the review could split the results into annual periods to check whether there is any evidence of trends over time [½]
- The results should be compared to standard tables for similar companies/industries to ensure that they are broadly consistent [½]
- The results should be brought up to date (e.g. by extrapolating past trends) [½]
- The results should be tested against the expected tables based on postcode review [½]
- In theory, medical information could be obtained for a sample of members to check whether there are any particular biases [½]

- The review should consider the extent to which IBNR deaths have been a feature in the past and an appropriate adjustment to the crude rates should be made [½]
 - The results should incorporate an allowance for future improvements to mortality [½]
 - Based on studies by insurance companies, the actuarial profession or other similar institutions [½]
- [Max 6]**

v) *The discounted cashflow approach*

The discounted cashflow approach was for many years the commonly used method for valuing schemes. Under the traditional presentation of this approach:

- Assets are valued by discounting the future expected cashflows based on a long term interest rate assumption. [1]
- Liabilities are valued using the same long-term interest rate assumption to discount the expected future liability outgo. [1]

The use of the same discount rate for valuing assets and liabilities ensures consistency and hence stability of the valuation result. The discounted cashflow approach has fallen out of favour with a trend towards schemes now adopting one of the market-related approaches. [1]

[Max 3]

- vi)** The stability of the valuation result could be argued to be artificial, and that a more appropriate valuation result should reflect current market conditions. [1]

The asset valuation method implicitly assumes stable asset cashflows that may not reflect the real world, *eg* income stream on equities. [1]

The discounted cashflow approach can be difficult to explain to clients, particularly the approach taken to value assets. [1]

Regulatory tests and modern accountancy standards use market-related approaches to asset valuation. [1]

[Max 4]

[25 Marks]

Solution 4:

i) Method of provision

If a benefit scheme is being discontinued, a number of options may exist for the provision of the outstanding benefit payments. Six examples are covered below:

- gradual removal of liabilities through temporary continuation [½]
- transfer the liabilities to another pension scheme with the same sponsor [½]
- transfer the funds directly to the beneficiary to extinguish the liability [½]
- transfer the funds to an insurer to invest and provide a benefit [½]

- transfer the liabilities to an insurer to guarantee the benefits [½]
- transfer the liabilities to a central discontinuance fund. [½]

[Max 3]**ii) A self-sufficiency basis**

- Adopts very prudent assumptions [½]
- Assumes a derisked investment strategy [½]
- That the scheme will continue to meet its liabilities (including expenses) on an ongoing basis [½]
- On the assumption that the employer will not provide any further financial support [½]

Scenarios include:

- The scheme is closed to accrual but the scheme does not intend to wind up. [½]
- The sponsor covenant is weak ie the sponsor is less able or willing to contribute to the scheme [½]
- The trustees are risk-averse and wish to invest in assets that best match the liabilities. [½]

[Max 5]**iii)**

- IBR tends to be most appropriate when there is some uncertainty about the sponsor's financial position, rather than as a regular monitoring tool. However, it can take explicit account of the interdependence of funding and the sponsor covenant and can help trustees to determine how much the sponsor can afford. [1]
- To reduce the costs associated with divesting and transferring the assets to an external provider [½]
- To avoid investing in insurance contracts which generate profits for the provider [½]
- There may be a lack of insurers willing to take on the liabilities [½]
- To retain freedom to invest in some higher yielding assets, such as corporate bonds [½]
 - Which may generate surplus assets in the future [½]
 - Which may allow the trustees to award discretionary benefits such as pension increases [½]
- There may still be a deficit on insurance company [½]
- Which the sponsoring company cannot afford to pay [½]
- The sponsoring company may be willing and able to stand behind the scheme [½]
- Should a deficit emerge in the future [½]
- The trustees may be looking for a more cost efficient means of managing longevity risk [½]
- Such as the use of a swap contract [½]
- As the scheme is fully funded on self-sufficiency the benefit security risk is likely to be within the trustees' risk tolerance [½]
- As self-sufficiency is likely to cost less than buy out the risk of members' benefits being reduced due to insufficient funding is lower if the self-sufficiency approach is retained. [½]

[Max 5]**iv)**

- The scheme will continue to run some investment risk [½]
- As it will be impossible to find assets that exactly match the cash outflows from the scheme [½]

- And longevity risk [½]
- Although this can be mitigated to a large extent through the use of a swap contract [½]
- There is a risk that the expenses involved in running the scheme are higher than expected in particular as the scheme shrinks in size. Therefore, it is possible that deficits will emerge in the future [½]
- So that the scheme will become reliant on the sponsor once again [½]
- And the ability and willingness of the sponsor to meet these unexpected payments at some unknown future point of time will be in doubt [½]
 - Therefore, the members continue to run the risk that benefits may not be paid in full [½]
- This risk is likely to impact the younger members to a greater extent since it is the later payments that are most at risk [½]
- The application of any future surplus may not be equitable across the membership [½]
- For example, some members may have died before the surplus has been generated [½]
- This risk is likely to impact the older members to a greater extent since they are more likely to die sooner [½]
- It is likely that it will become uneconomic to run the scheme in this way at some point in the future [½]
- So that the trustees will probably need to approach the insurance market eventually [½]
- And the terms then on offer are unknown and may have deteriorated (e.g. as a result of potential excess demand from schemes) [½]

[Max 6]

- v) For pension schemes in general, the liabilities have a very long duration, often greater than the duration of the assets held, leading to interest rate risks. [½]

The long duration of pension scheme liabilities can make it difficult to choose an investment strategy that matches the liabilities well. Matching by duration can be prohibitively expensive because, for example, bonds of a suitably long term may be in short supply. [½]

However, once a defined benefit scheme closes to future accrual the duration of the liabilities and so the level of interest rate risk will gradually reduce. Many strategies involve swaps. Swaps provide a relatively straightforward vehicle for schemes to control the amount of mismatch in timing or maturity of the assets and liabilities. [½]

A swap can be tailored so that the payments received by the scheme closely match the scheme's pension payments. It is generally possible to match these pension payments more closely with swaps than with a portfolio of fixed-interest bonds. [½]

Under a typical swap contract, the scheme will receive fixed payments for the duration of the contract in exchange for floating payments, the floating payments often being linked to an index. [½]

Such a strategy will help to hedge interest rate risk for the scheme. For example, if long-term interest rates fall, then:

- the discount rate may reduce and the value placed on the scheme's liabilities will increase
- but the value of the scheme's assets will also increase because the value of the swap to the scheme will increase (as the scheme will still receive the same fixed payments but will now make smaller floating payments as interest rates are lower). [½]

Longevity swaps and longevity bonds

A major risk affecting defined benefit pension schemes is longevity risk. This can be mitigated by investing in longevity bonds or swaps. [½]

Longevity bonds work where the coupon payments are linked to the mortality experience of a particular set of lives, such as a specific age group in the national population. Coupon payments will reduce over time in line with the mortality experience of that population. [½]

Larger schemes may be able to arrange a longevity swap with an insurance company where the payments are linked to the mortality experience of the scheme's membership rather than the national population. [½]

If longevity is higher than expected, the coupon payments received will be higher, thus hedging the longevity risks of the pension scheme. The hedge is not perfect if the population underlying the bond is not the same as that in the pension scheme. [½]

For example, if the pension scheme members live longer on average than the population being tracked. The hedge will also not be perfect if investment returns and / or pension increases are at a different rate than expected. [½]

The market for longevity bonds is currently very small. [½]

[Max 6]
[25 Marks]
